ahmad_mahmoud.resume.ts FULL-STACK ENGINEER ● 12+ YRS

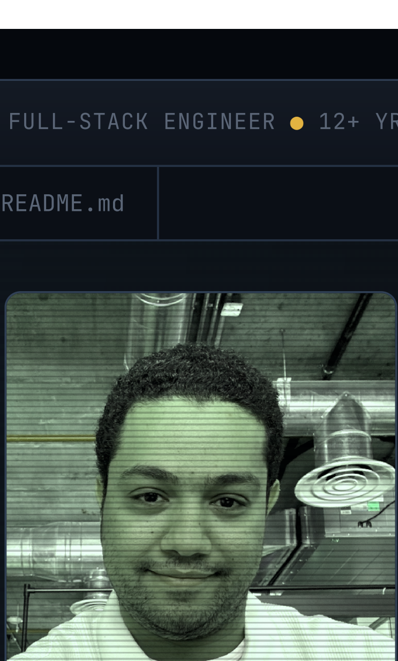
● full_stack.ts | experience.json | projects/ | stack.config | README.md

→ whoami --full

Ahmad Abdul-Jawad Mahmoud

Full-Stack Engineer / Backend · Web · Mobile

// **backend** · **web** · **mobile** · **12+ years** in production · Cairo ⟶ Riyadh · database to App Store



4	320+	2,900+	12+
LAYERS BUILT SOLO	API ENDPOINTS	BACKEND TESTS	YEARS ENGINEERING

```
/**
* Full-Stack Engineer — I own a feature from the database to the
* app on the phone. Right now: a production ride-hailing platform built solo
* across every layer — a NestJS / TypeScript backend on PostgreSQL and Redis
* (23 modules, ~320 endpoints, 2,900+ tests), a Next.js ops console in
* production, realtime Socket.IO feeds, payments, and native iOS & Android
* apps on one API contract.
* Behind it: 12+ years of production engineering, seven of them owning a
* POS used by 20k+ restaurants.
*/
```

## across_the_stack()

4 pillars

[01] backend & data

Correct before clever

A NestJS modular monolith (**23 modules**) on PostgreSQL with versioned migrations; Redis for dispatch state, the driver geo index, rate limits and cron leases. A **double-entry ledger**, a transactional outbox and idempotency keys, so money and side effects survive retries.

◆ 23 modules · ledger · outbox

[02] realtime & integrations

The parts that talk to the world

Socket.IO gateways for riders, drivers and admins; Google Maps routing with traffic-aware fare quotes and caching; Paymob cards, Apple Pay and Google Pay; WhatsApp-first OTP with SMS fallback; APNs and FCM push.

◆ Sockets · maps · payments · OTP · push

[03] web

An ops console people can run a business on

A **30-screen** Next.js 14 console: live MapLibre trips map, dispatch and SOS boards, KYC review, finance reconciliation, refunds and payouts, a map zone editor, feature flags, **RBAC** and an audit log behind Google SSO.

◆ 30 screens · RBAC · audit log

[04] mobile clients

12 years of shipping apps

Native SwiftUI (Swift 6) and Kotlin / Compose apps on shared, versioned core libraries, with **800+ iOS tests**. Before this: apps used by **35M+ people** — HungerStation, Ego ride-hailing and the Marn POS.

◆ SwiftUI · Compose · 800+ iOS tests

## experience()

8 roles

Full-Stack Engineer — Backend · Web · iOS · Android @

Ride-Hailing Platform · Independent — Mar 2026 → Present CURRENT

Took the **full technical scope of a founding CTO** and carried it alone — building a production ride-hailing platform **solo**: a hardened backend and a real-time ops console **in production**, with native **iOS/SwiftUI** and **Android/Compose** apps (Rider & Driver on each) in active development — **owning every layer** and setting the engineering standards (idempotency, double-entry accounting, acyclic module boundaries, CI test suite).

› **Backend APIs**
// NestJS · TypeScript · Mar '26 → now

Modular monolith, **23 modules** — **trip state machine**, **scored dispatch** over a Redis geo index, **double-entry ledger**, realtime rider/driver/admin **sockets**, JWT + phone-OTP, idempotency & audit log · ~320 endpoints over 47 controllers, traffic-aware fare quotes, **2,900+ tests**, Swagger docs.

› **Admin Console**
// Next.js 14 · Jun '26 → now

Bespoke **30-screen** ops surface — live **MapLibre** trips map, dispatch/SOS live boards, KYC onboarding review, finance reconciliation, refunds/payouts, cash control, map **zone editor**, feature flags, audit log, **Google SSO**.

› **iOS Apps · Rider & Driver**
// SwiftUI · Jul '26 → now

Two native **SwiftUI** apps on a shared **BuraqCore** Swift package (domain · networking · design system, pinned remotely) — live GPS **trip lifecycle**, Socket.IO feeds with fresh-token reconnect, in-app chat, payments, Live Activities, driver earnings & turn-by-turn navigation · Swift 6 strict concurrency, **800+ tests**.

› **Android Apps · Rider & Driver**
// Kotlin · Jetpack Compose · Sep '26 → now

Native **Kotlin / Compose** ports of both apps against **the same API contract**, on a shared **buraq-core-android** library published to GitHub Packages — driver KYC onboarding, offers and trips; rider booking, live tracking, FCM push and Google Pay · **290+ test suites**.

One backend · one contract · four native clients in flight — plus growth loops (promotions, referrals, loyalty, wallet), driver safety/SOS, and scheduled reconciliation jobs, owned end-to-end.

iOS Team Lead @ Marn — Jahez Group

Aug 2019 → Aug 2026

RIYADH, KSA

- **Led a 5+ engineer iOS team** for seven years — onboarding, mentoring, career growth, and the review culture behind an enterprise POS used by **20k+ restaurants** and cafés.
- **Owned the technical roadmap and architecture:** an offline-first CoreData core with cloud sync, built so a venue keeps trading when the network does not.
- **Set the engineering standards** — module boundaries, code review, automated testing — holding the app at a **99.8–99.9% crash-free** rate in production.
- **Drove the multi-year migration** to SwiftUI, Combine and async/await without pausing feature delivery, and made the case for it to product leadership.
- **Built CI/CD and the release process from scratch**, cutting deployment time **60%** and shortening every engineer's feedback loop.
- Ran the performance programme (**40% faster load times**) and designed the table-management, order-processing and inventory systems at the product's core.
- Partnered with product, design and backend leadership on scope, estimates and delivery commitments.

Senior iOS Engineer @ HaramBlur · voluntary charity project

Dec 2025 → Jun 2026

- Designed an architecture integrating **on-device machine learning** for real-time content analysis.
- Built a seamless communication layer between native iOS and web-based content.
- Implemented efficient image processing with region-based and full-image blur effects.
- Created a real-time video monitoring system for continuous content filtering.
- Developed smart banner blocking and optimised memory management throughout.

iOS Team Lead @ Ego — Ride Hailing

Mar 2018 → May 2019

RIYADH, KSA

- Developed a ride-hailing ecosystem (iOS, watchOS) serving **15M+ users** in Saudi Arabia.
- Implemented real-time GPS tracking and navigation with MapKit and optimised route algorithms.
- Integrated Apple Pay, HyperPay, and push-notification infrastructure.
- Created seamless iPhone ⟷ Apple Watch handoff for continuous ride tracking.
- Built the driver rating system and real-time fare calculation engine.

Senior iOS Engineer @ HungerStation — DeliveryHero

Sep 2016 → Mar 2018

KSA

- Contributed to a major food-delivery platform serving **20M+ users**.
- Optimised app performance for high-volume transaction processing.
- Implemented real-time order tracking and delivery optimisation.
- Collaborated with backend teams on API design and integration.

iOS Developer @ ALNapedh

May 2016 → Aug 2016

- Developed food-ordering apps with Firebase integration and real-time features.
- Implemented Spreedly payments and live-chat functionality.
- Built responsive UI with Auto Layout across multiple device sizes.

iOS & Apple Watch Developer @ Nile Design Co.

Jul 2015 → Sep 2015

- Early adopter of Apple Watch development (watchOS 1–2).
- Built a foundation in iOS and watchOS app development.
- Adopted SwiftUI from WWDC 2019 onward.

iOS Developer @ UK Pro. Solutions Ltd.

May 2014 → May 2015

- Foundation iOS experience in Objective-C and early Swift.
- Built social networks and e-commerce platforms.
- Learned core iOS principles and App Store deployment.

## projects[]

selected

[01] ai / privacy

HaramBlur — Privacy Browser

SwiftUI · CoreML · WKWebView · CoreImage · Vision

Privacy-focused iOS browser with real-time, AI-powered content detection and filtering. On-device ML, smart blur, live video monitoring.

◆ AI-powered privacy browser

[02] full-stack

Ride-Hailing Platform

SwiftUI · Kotlin · Jetpack Compose · NestJS · TypeScript · PostgreSQL · Redis · Socket.IO · Next.js

End-to-end ride-hailing system built solo: a hardened NestJS backend (idempotency, double-entry ledger, 2,900+ tests) and a real-time Next.js ops console in production, with **four native clients** (Rider & Driver on iOS and on Android) in active development.

◆ Full-stack ownership · prod security

[03] delivery

HungerStation

UIKit · Firebase · Real-time APIs · Perf

Major food-delivery platform serving 20M+ users across Saudi Arabia and Bahrain. Performance and real-time order tracking.

★ 4.5 / 5 — 1.9M+ ratings

[04] mobility

Ego — Ride Hailing

MapKit · Core Location · Apple Pay · watchOS

Complete ride-hailing ecosystem: rider, driver, and Apple Watch apps. Real-time GPS and cross-device experiences for 15M+ users.

★ 4.7 / 5 — 211.8k+ ratings

[05] enterprise

Marn — iPad POS

SwiftUI · CoreData · CloudKit · Real-time Sync

Enterprise POS serving 20k+ restaurants across Saudi Arabia. Offline-first architecture, complex table management, real-time orders.

★ 4.8 / 5 — 2,437 ratings

[06] community

Daleelo

MapKit · Core Location · Social SDKs

Muslim community app — prayer times, mosque finder, and community features with enhanced location + social integration.

★ 4.5 / 5 — 1,287 ratings

## stack.config

web_and_infrastructure =
Next.js 14 · React · TypeScript · MapLibre · Socket.IO · Docker · GitHub Actions CI · Prometheus Metrics · Jest · Google Maps Platform · Paymob · Cloudflare R2

ios_2025 =
Swift 6.0 · SwiftUI · UIKit · Combine · async/await · SwiftData · iOS 18+

android_2025 =
Kotlin · Jetpack Compose · Android · Coroutines · Flow · MVVM

architecture =
MVVM · VIPER · Clean Architecture · Coordinator · TDD

apple_ecosystem =
watchOS · tvOS · macOS · WidgetKit · Live Activities · App Intents

intelligence =
Core ML · Vision · ARKit · Dynamic Island · App Tracking

backend_apis =
NestJS · TypeScript · PostgreSQL · TypeORM · Redis · JWT/OTP · REST · GraphQL · Firebase · CloudKit · WebSockets · Swagger

tooling =
Xcode 15+ · Fastlane · TestFlight · Instruments · Git · Copilot · Cursor

## portfolio/

FOOD & GAMING
- HungerStation
- Punters

TRANSPORTATION
- Ego — Rider
- Ego — Driver
- Ego — Watch

ENTERPRISE
- Marn POS
- Medical Guide

SOCIAL & COMMUNITY
- Daleelo
- NU-inc
- VeeU

WATCHOS
- Ego Watch
- TaQs Weather
- TweetMe

E-COMMERCE
- iBss
- Exclusive Luxe
- Scratch

RELIGIOUS & EDU
- Afasy
- ArabNileGroup

UTILITY & TOOLS
- Unlocking Smart
- Insta
- iContacts
- First iOS app

AI & ML
- HaramBlur
- RecognizeMe
- Grad project · OpenCV

WEB
- FaceDemo
- Web Pages

## record{}

// education

B.Sc. Information Systems
Computer Sciences College
Asyut University, Egypt · 2009–2013
iOS Semester — Excellent (91.5%)

// certification

ISTQB Foundation Level
Egyptian Software Testing Qualification Board

// languages

Arabic	native
English	professional

## honours[]

01 Building a **production ride-hailing platform solo across every layer** — backend, ops console, and iOS and Android apps on one API contract.

02 Held a **99.8–99.9% crash-free** rate and cut deployment time **60%** with the standards, tests and CI/CD the team ran on.

03 Architecting a **full-stack ride-hailing platform solo** — NestJS/PostgreSQL backend and Next.js ops console in production, four native clients (rider and driver, iOS and Android) in development.

04 Built a food-delivery platform serving **20M+ users** and a ride-hailing ecosystem serving **15M+ users** in Saudi Arabia.

05 **10+ apps** on the App Store, used by millions across MENA.

06 **30+ projects** delivered across a 12-year career.

07 Early Swift adopter (WWDC 2014) & early SwiftUI adopter (WWDC 2019).

main | ● Full-Stack Engineer · Backend · Web · Mobile | Cairo ⟶ Riyadh | UTF-8 | TypeScript · Swift · Kotlin | ★ 12+ yrs · 4 layers solo